

Curriculum Vitae

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Education

2022 - Present	Ph.D., Physics University of Washington Seattle, WA 98195
2017 - 2021	B.A., Physics + Economics University of Pennsylvania - Mayor's Scholar 2017-2021 Philadelphia, PA 19104

Research Interests

- Differentiable simulation
- Generative simulation of high-energy particle detectors
- Quantum information
- Hardware-accelerated ML for real-time inference

Publications

1. A. Elabd, E. Gross, D. Kobylanski, B. Nachman, “Parnassus: A GPU-enabled, Python-based Package for Fast Particle Detector Simulation and Reconstruction,” arXiv:2606.26089 [hep-ex] (2026). 🔗 [arXiv](#)
2. A. Elabd *et al.*, “Graph Neural Networks for Charged Particle Tracking on FPGAs,” *Front. Big Data* **5**, 828666 (2022). 🔗 [DOI](#)

Awards and Funding

2025-2027	WATCHEP Fellowship
2021	IRIS-HEP Fellowship
2017-2021	Mayor's Scholarship

Invited Talks

May 2026	<i>Adapting Parnassus: A Fast-Simulation Tool for the ATLAS Experiment</i> Computational High-Energy Physics (CHEP) 2026, Bangkok, Thailand
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Contributed Talks

Sep 2025	<i>FastML for Real-Time RHEED Inference</i> Fast Machine Learning for Science 2025, Zurich, Switzerland
May 2022	<i>Graph Neural Networks for Charged Particle Tracking on FPGAs</i> Connecting the Dots 2022, Princeton, NJ

Blogs and Outreach

- **Founder:** UW Quantum Computing Hardware Journal Club
- **Instructor:** AI Future Lab, U. Washington, 2026
- **Lead Hackathon Organizer:** U. Washington chapter of [HDR FAIR ML Challenge 2025/2026](#)
- **Blog Author:** [A3D3 at FastML 2025](#)
- **Instructor:** [A3D3 AI Summer Camp 2025](#)
- **Instructor:** Quarknet CMS Masterclass, U. Washington: 2022 - 2026

Research Experience

- Aug 2025 - Present **Search for Quantum Entanglement and Bell Nonlocal States in $Z \rightarrow \tau\tau$**
- Use ML and Monte Carlo to quantify quantum “magic” in $Z \rightarrow \tau\tau$ decays at the LHC.
 - Fine-tune foundation model to perform neutrino/hidden-particle reconstruction.
- Mar 2025 - Present **WATCHEP Fellow**
- [Western Advanced Training in Computational High-Energy Physics](#) (WATCHEP).
 - Developed [TorchDelphes](#): a differentiable version of Delphes simulation software
 - Implementing gradient-based tuning of TorchDelphes on CMS full-simulation data
 - Developing iterative, likelihood-based algorithm to unfold high-energy physics data with generative ML.
 - Deploy parallel, multi-node training on Perlmutter supercomputer at LBNL/NERSC.
- Feb 2025 - Present **ATLAS Pixel Detector DAQ development**
- FPGA development for ATLAS Pixel Detector data-acquisition (DAQ) system.
 - Develop, simulate, deploy co-processor FPGA designs with embedded Linux kernels.
- Sep 2024 - Present **Real-time inference on FPGAs for thin-film deposition with RHEED**
- Lead software-hardware codesign of ML inference chain for real-time characterization of crystalline material synthesis via thin-film deposition.
 - Supervise ML workflow including data augmentation and simulation, self-supervised learning, neural pruning and architecture search, and quantization-aware training.
- Jun - Sep 2024 **Reinforcement learning for accelerated calibration of trapped-ion qubits**
- Trained [reinforcement learning \(RL\) agent](#) to calibrate two-qubit entangling gates using sparse, stochastic, (quantum-)measurement-based adaptive feedback.
 - Developed [noise-model simulation](#) of Ca-40⁺ optical trapped-ion qubits.
- Mar - Jun 2024 **AI- and hardware-accelerated readout of superconducting qubits**
- Implemented, tested FPGA ML algorithms for scalable, low-latency readout of superconducting qubits with the open-source [QICK](#) RFSoc qubit controller.
- Jan 2021 - Apr 2022 **IRIS-HEP Fellow: Graph Neural Networks for Charged Particle Tracking on FPGAs**
- Developed [hls4ml.pyg_to_hls](#) toolkit for automatic translation of Graph Neural Networks (GNNs) into optimized High-Level Synthesis (HLS) FPGA designs.
 - Completed feasibility study on use of FPGA-deployed GNNs for sub-microsecond charged-particle tracking at the CMS Level 1 trigger.

Industry Experience

- Aug 2021 - Oct 2023 **Jr. Algorithm Engineer**
- TDI Technologies, Inc., Philadelphia, PA
- Developed, deployed ML algorithms for systems monitoring aboard US Navy vessels.
 - Provided documentation and testing instructions for enlisted personnel, engineers.
 - Security clearance: Secret.

Computation

- **Full-stack FPGA:** (System)Verilog, VHDL, HLS, hls4ml, Vivado, ModelSim
- **Software:** Python, C++, Julia, Mathematica, Linux, Git
- **Reinforcement Learning:** OpenAI Gym
- **Machine Learning:** PyTorch, TensorFlow, scikit-learn
- **High-Performance Computing:** SLURM, Horovod, torch.DDP, Lightning
- **Numerical simulation**
- **Quantum Computing:** QICK, Azure Quantum, qiskit, QuTiP

Conferences and Development

May 2026	Computational High-Energy Physics 2026 Chulalongkorn University, Bangkok, Thailand
Sep 2025	Fast Machine Learning for Science 2025 ETH, Zurich, Switzerland
Aug 2025	A3D3 All-Hands Meeting California Institute of Technology, Pasadena, CA
June 2025	WATCHEP Summer School U. California, Berkeley, CA
April 2025	ATLAS Pixel Week CERN, Geneva, Switzerland
March 2025	Deep Learning at Scale NERSC, Berkeley, CA
Sep 2024	NSF Harnessing the Data Revolution (HDR) Ecosystem Conference University of Illinois at Urbana-Champaign, Champaign, IL
Jul 2024	US Quantum Information Science Summer School Oak Ridge National Laboratory, Oak Ridge, TN
May 2022	Connecting the Dots 2022 Princeton University, Princeton, NJ

Teaching

Winter 2025	PHYS522 Implementations in Quantum Computing
Autum 2024	PHYS521 Quantum Information Theory
Autum 2024	PHYS433 Nuclear and Particle Laboratory
Autumn 2023	PHYS434 Computational Data Analysis
Spring 2023	PHYS417 Neural Network Methods for Signals
(Multiple)	PHYS334 Advanced Laboratory Electronics
(Multiple)	PHYS232 Computational Physics

Notable Coursework

Winter 2026	PHYS567	Solid State Physics
Winter 2025	EE271	Intro to FPGAs
Winter 2024	PHYS576B	Implementations in Quantum Computing
Autumn 2023	PHYS521	Quantum Information Theory
Spring 2023	PHYS564	General Relativity
Spring 2021	ECON224	Statistical Learning and Causal Inference
Spring 2021	ECON212	Game Theory
Spring 2020	PHYS585	Theoretical and Computational Neuroscience
Autumn 2019	STAT430	Probability
Spring 2019	PHYS364	Electronics and Digital Design
Spring 2018	LING105	Intro to Formal Linguistics